CompSci Graph Theory Notes

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Outline

Computer Science and Graph Theory

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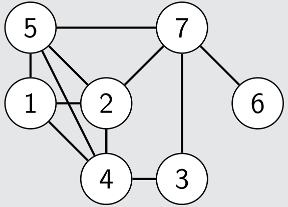
Representing Graphs

- Graphs can be represented using adjacency matrices
- Adjacency matrices are rectangular arrays that use 1 to indicate edges in a graph

Example (Finding an adjacency matrix)

Find the adjacency matrix for the following graph:



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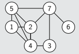

Matrices and Python

- Matrices can be handled in Python using the NumPy `arrays`:

```python
import numpy as np

A = np.array([
    [0, 1, 0, 1, 1, 0, 0],
    [1, 0, 0, 1, 1, 0, 1],
    [0, 0, 0, 1, 0, 0, 1],
    [1, 1, 1, 0, 1, 0, 0],
    [1, 1, 0, 1, 0, 0, 1],
    [0, 0, 0, 0, 0, 0, 1],
    [0, 1, 1, 0, 1, 1, 0]
])
```


Matrices and Python

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import numpy as np

A = np.array([
    [0, 1, 0, 1, 1, 0, 0],
    [1, 0, 0, 1, 1, 0, 1],
    [0, 0, 0, 1, 0, 0, 1],
    [1, 1, 1, 0, 1, 0, 0],
    [1, 1, 0, 1, 0, 0, 1],
    [0, 0, 0, 0, 0, 0, 1],
    [0, 1, 1, 0, 1, 1, 0]
])

Operations on Arrays

- NumPy provides many operations to use with arrays
  1. Column/row sums with `sum` (*not* `np.sum`). This is useful for finding degrees of vertices.
  2. Matrix multiplication with `@` (such as `A@A`). This is useful for counting paths between vertices.

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  1. Column/row sums with `sum` (*not* `np.sum`). This is useful for finding degrees of vertices.
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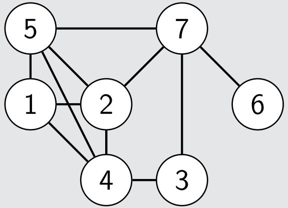
Example (Adjacency matrices, paths and degrees)

Use the adjacency matrix to answer the following questions:

1. What is the degree of each vertex?
2. How many paths of length two are there between vertex 4 and vertex 7?
3. How many cycles of length four start at vertex 2?



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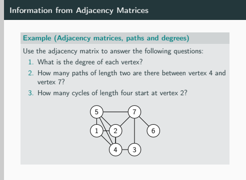
CompSci Graph Theory Notes
└─Computer Science and Graph Theory
└─Information from Adjacency Matrices



Drawing Graphs

- We can draw graphs using NetworkX and matplotlib.

```python
import matplotlib.pyplot as plt
import networkx as nx

G = nx.from_numpy_array(A)

nx.display(G)
plt.show()
```

```
None
```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```python
import matplotlib.pyplot as plt
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```

None

This code should be continued from previous code!

Using the algorithm in Python

Using the provided algorithm `gale_shapley_algorithm.py`, solve the stable matching problem for the following set of preferences:

Applicant	Employer rankings	Employer	Applicant rankings
1	B, A, C	A	3, 1, 2
2	B, C, A	B	2, 3, 1
3	A, B, C	C	1, 3, 2

Gale-Shapley Algorithm
Using the algorithm in Python
Using the provided algorithm gale_shapley_algorithm.py, solve the stable matching problem for the following set of preferences:
Applicant Employer rankings Employer Applicant rankings
1 B, A, C A 3, 1, 2
2 B, C, A B 2, 3, 1
3 A, B, C C 1, 3, 2